Digital Egypt Pioneers Initiative

Smart Fleet – Intelligent Fleet & Delivery Management Platform

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Stakeholder Analysis – SmartFleet

Primary stakeholders

Dispatcher
Creates and assigns deliveries and monitors them until completion.
Needs: a fast way to choose the right vehicle and driver, a clear view of every delivery's status, and instant alerts when something changes.
Influence: High | *Interest: High*

Driver
Executes the deliveries on the road.
Needs: a clear list of his assigned deliveries, a simple way to update status, and an easy way to log fuel and expenses on the spot.
Influence: Medium | *Interest: High*

Fleet Manager
Responsible for the condition and efficiency of the whole fleet.
Needs: up-to-date vehicle and driver records, maintenance scheduling with alerts, and dashboards for utilization, maintenance costs, and fuel expenses.
Influence: High | *Interest: High*

Admin
Controls access and keeps the system secure.
Needs: full control over users and roles, a secure login mechanism, and visibility over overall performance.
Influence: High | *Interest: Medium*

Secondary stakeholders

Company management / Operations Director
Sponsors the system and judges it by results.
Needs: lower operating costs, on-time deliveries, and reliable analytics to make decisions.
Influence: High | *Interest: Medium*

Development team
Builds, tests, and maintains the system.
Needs: clear and stable requirements, a clean and scalable architecture, and security and testing built in from the start.
Influence: Medium | *Interest: High*

Customers receiving the deliveries (indirect)
They do not use the system, but they feel its effect.
Needs: deliveries that arrive on time and reliably.
Influence: Low | *Interest: Low*

User story :

Smart Fleet is a centralized platform that helps a company manage its vehicles, drivers, deliveries, maintenance, fuel, and expenses in one place. Everything starts when someone opens the system.

As a dispatcher, driver, fleet manager, or admin, I want to log in securely with my own account, so that only authorized people can access the system and each of us sees only what our role allows.

Once logged in, the day begins with the dispatcher, who is responsible for getting deliveries out the door. As a dispatcher, I want to create a new delivery and assign it to a vehicle and a driver, so that every delivery has clear resources allocated to it. Choosing the right vehicle is the hardest part, because capacity, availability, maintenance status, and driver availability all matter at once. So, as a dispatcher, I want the system to recommend the most suitable vehicle for each delivery based on availability, capacity, maintenance status, driver availability, fuel consumption, and utilization, so that I can assign deliveries faster and make better decisions.

After the assignment, the delivery moves on to the driver, who needs to know what is expected of him. As a driver, I want to view the deliveries assigned to me, so that I know my tasks without waiting for a phone call. And as I progress through the trip, I want to update the status of my delivery, so that everyone knows where it stands.

While the driver is on the road, the dispatcher needs the full picture. As a dispatcher, I want to track the status of all deliveries, so that I can react quickly when something is delayed, and I want to update a delivery's status when needed, so that the records stay accurate even when the driver cannot do it. To keep both of us in sync without constant phone calls, as a dispatcher or driver, I want to receive real-time notifications when a delivery or vehicle status changes, so that we always work with the latest information.

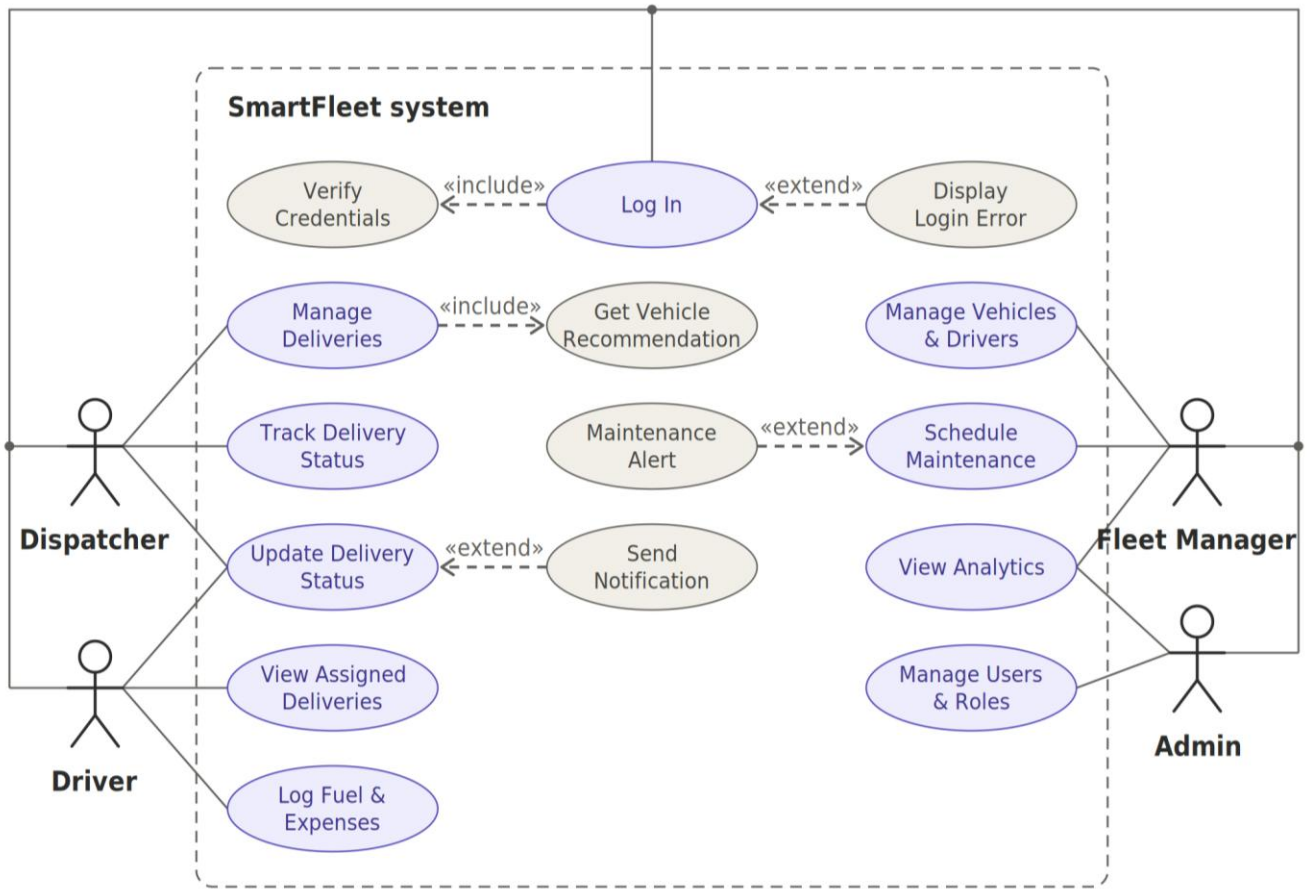
Every trip also costs money, and that must be recorded while it is fresh. As a driver, I want to log fuel and expenses, so that costs are recorded accurately and nothing gets lost.

Behind the daily operation, the fleet manager keeps the fleet itself in good shape. As a fleet manager, I want to manage vehicles and drivers, so that the fleet data is always complete and up to date. I also want to schedule maintenance and be alerted when it is due, so that no vehicle is serviced late or sent out when it should not be.

Over time, all this activity produces data, and management needs to turn it into decisions. As a fleet manager, I want to view dashboards on fleet utilization, maintenance costs, fuel expenses, and delivery performance, so that I can understand operational trends and make informed decisions. As an admin, I want to see the same dashboards, so that I can monitor the overall performance of the operation.

Finally, someone must control who is allowed to do what. As an admin, I want to manage users and their roles, so that every person has only the permissions they need and the company’s data stays secure.

Use-case:


SmartFleet system
Verify Credentials
«include»
Log In
«extend»
Display Login Error
Manage Deliveries
«include»
Get Vehicle Recommendation
Manage Vehicles & Drivers
Track Delivery Status
Maintenance Alert
«extend»
Schedule Maintenance
Update Delivery Status
«extend»
Send Notification
View Analytics
View Assigned Deliveries
Manage Users & Roles
Log Fuel & Expenses
Dispatcher
Driver
Fleet Manager
Admin

Functional Requirements

Authentication & Access Control

- The system shall allow dispatchers, drivers, fleet managers, and admins to log in with their own accounts.
- The system shall verify the user's credentials and display an error message when the login fails.
- The system shall restrict each user to the features allowed by their role.
- The system shall allow admins to create users and assign or change their roles.

Delivery Management

- The system shall allow dispatchers to create deliveries and assign each one to a vehicle and a driver.
- The system shall allow dispatchers to track the status of all deliveries.
- The system shall allow dispatchers and drivers to update a delivery's status.
- The system shall allow drivers to view the deliveries assigned to them.

Vehicle Recommendation

- The system shall recommend the most suitable vehicle for each delivery.
- The recommendation shall consider vehicle availability, capacity, maintenance status, driver availability, fuel consumption, and utilization.

Fleet Management

- The system shall allow fleet managers to add, edit, and remove vehicles and drivers.

- The system shall show the availability of vehicles and drivers.

Maintenance Management

- The system shall allow fleet managers to schedule maintenance for vehicles.
- The system shall alert fleet managers when maintenance is due.

Fuel & Expense Tracking

- The system shall allow drivers to log fuel consumption and expenses.
- The system shall store these records and link them to the related vehicle and delivery.

Notifications

- The system shall send real-time notifications to dispatchers and drivers when a delivery or vehicle status changes.

Dashboards & Analytics

- The system shall provide fleet managers and admins with dashboards showing fleet utilization, maintenance costs, delivery performance, fuel expenses, and operational trends.

Non-functional Requirements

Performance

- The system shall respond to common user actions (such as viewing deliveries or updating a status) quickly enough to be used comfortably during daily operations.
- Real-time notifications shall reach users with minimal delay.
- Dashboards shall load efficiently even as the volume of data grows.

Security

- The system shall authenticate all users and enforce role-based authorization on every feature.
- User passwords and sensitive data shall be stored securely and never exposed in plain text.
- All communication between the client and the server shall be encrypted.
- The system shall protect itself against common web attacks such as SQL injection and cross-site scripting.

Usability

- The interface shall be simple and consistent, so each role can perform its main tasks with little training.
- The interface shall work well on both desktop and mobile screens, since drivers use it on the road.
- Error messages shall be clear and tell the user what to do next.

Reliability

- The system shall be available during working hours, with minimal downtime.
- Data shall not be lost: records of deliveries, maintenance, fuel, and expenses shall be backed up regularly.
- The system shall handle failures gracefully and keep data consistent.

Scalability

- The system shall support growth in the number of vehicles, drivers, deliveries, and users without redesign.
- The system shall be deployable on Azure so resources can scale with demand.

Maintainability & Testability

- The system shall follow a clean architecture with clear separation between layers, so features can be changed or added easily.
- The system shall include automated tests for its core business logic.